

Report of Findings

Step 1: Dataset Overview (.info(), .describe())

- **Age, Cabin, and Embarked** had missing values.
- Used **median** to fill Age and **mode** to fill Embarked.
- **Cabin** was mostly null and ignored in the analysis.
- .describe() showed:
 - Average Age: ~29.7
 - Most passengers in Pclass 3
 - Fare is highly varied, some very high
 - info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

- describe()

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Step 2: Value Counts (Categorical Data)

- **Survived:** 38% survived, 62% did not.
- **Sex:** More males (577) than females (314).
- **Pclass:** Majority were in 3rd class (491).
- **Embarked:** Most passengers boarded from S (Southampton).

```
Survived:
Survived
0      549
1      342
Name: count, dtype: int64

Pclass:
Pclass
3      491
1      216
2      184
Name: count, dtype: int64

Sex:
Sex
male      577
female    314
Name: count, dtype: int64

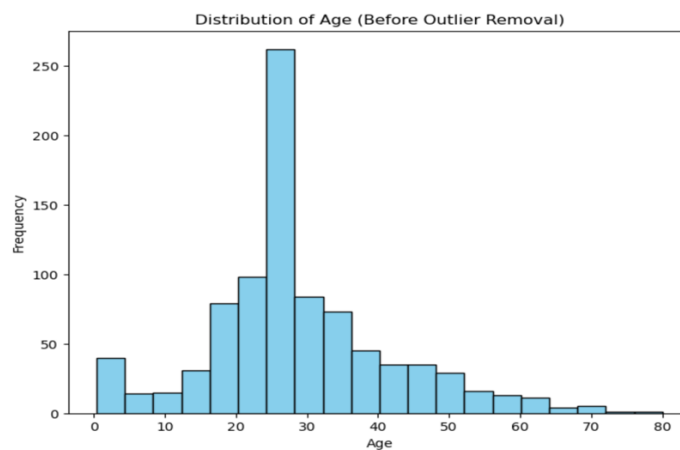
Embarked:
Embarked
S      646
C      168
Q       77
Name: count, dtype: int64
```

Step 3: Histograms (Age and Fare)

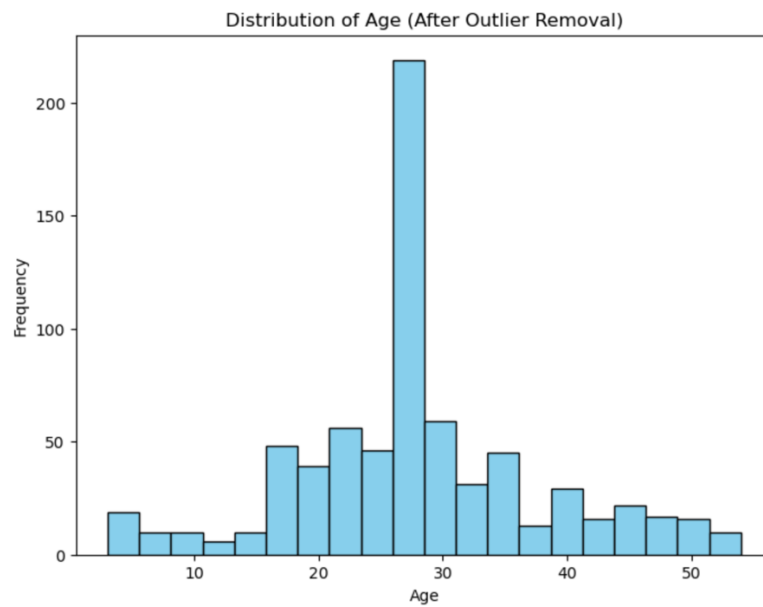
📌 Age Histogram

- Bell-shaped distribution, centered around 20–40 years.
- Very few passengers above 60.
- Implies majority were young adults.

Before Outlier Removal

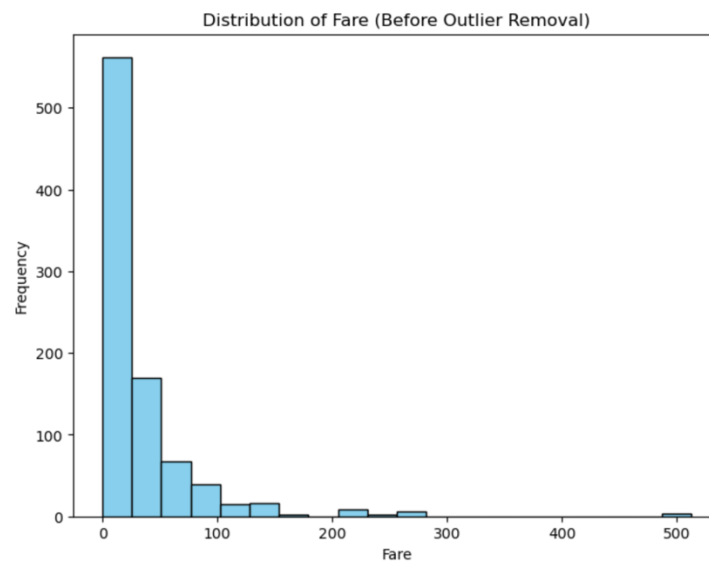


After Removal

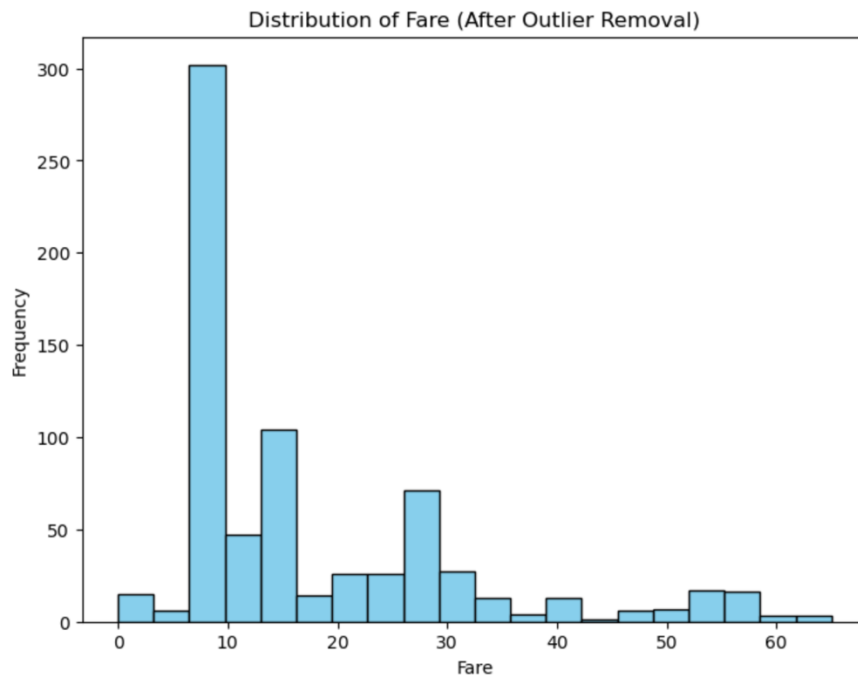


Fare Histogram

- Right-skewed with many fares below 100.
- Few outliers paid more than 300–500.
- Indicates possible **luxury tickets** for 1st class.
- Before Outlier Removal



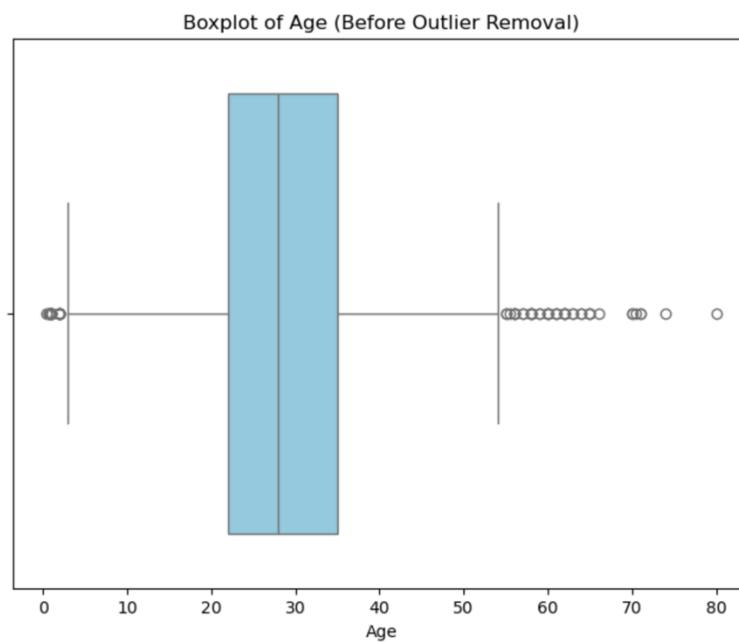
After Removal



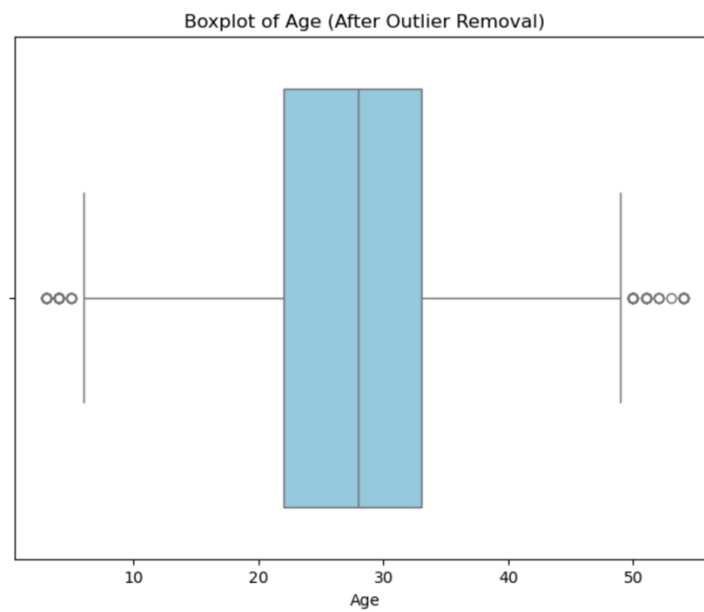
Step 4: Boxplots (Age and Fare)

📌 Age Boxplot

- Outliers present above age 65.
- Interquartile range mainly between 20 and 40.
- Before Outlier Removal



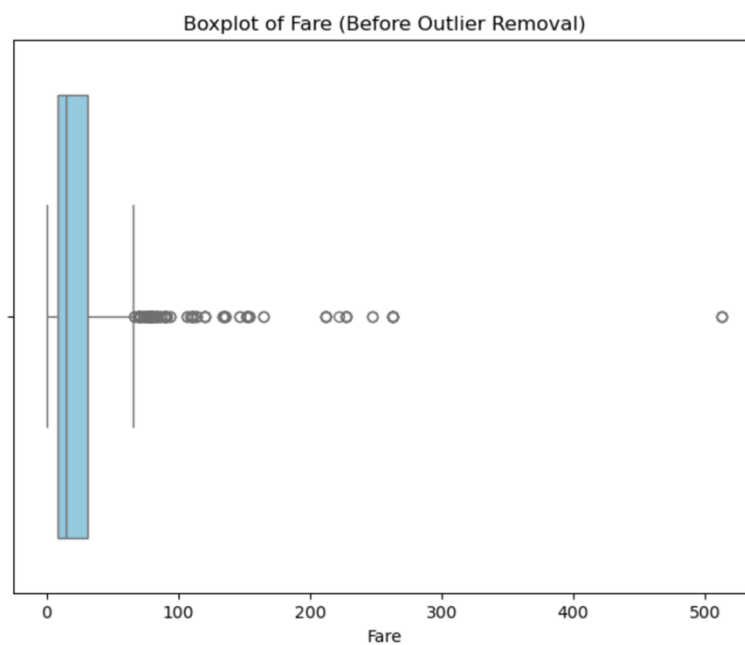
- After Removal



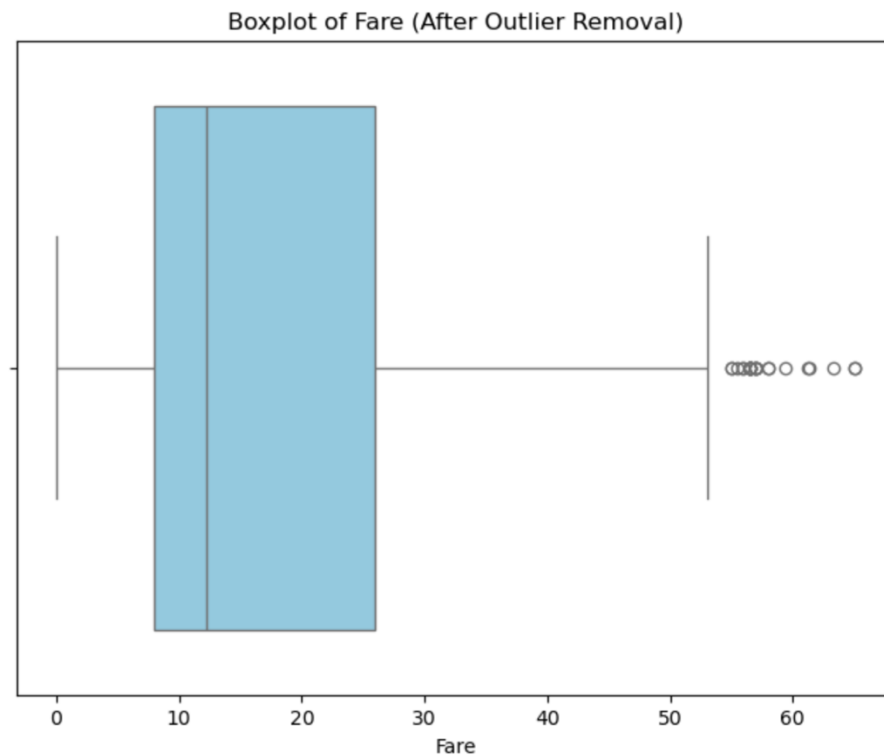
Fare Boxplot

- Extreme outliers above 300.
- Most fares below 100.

Before Removal

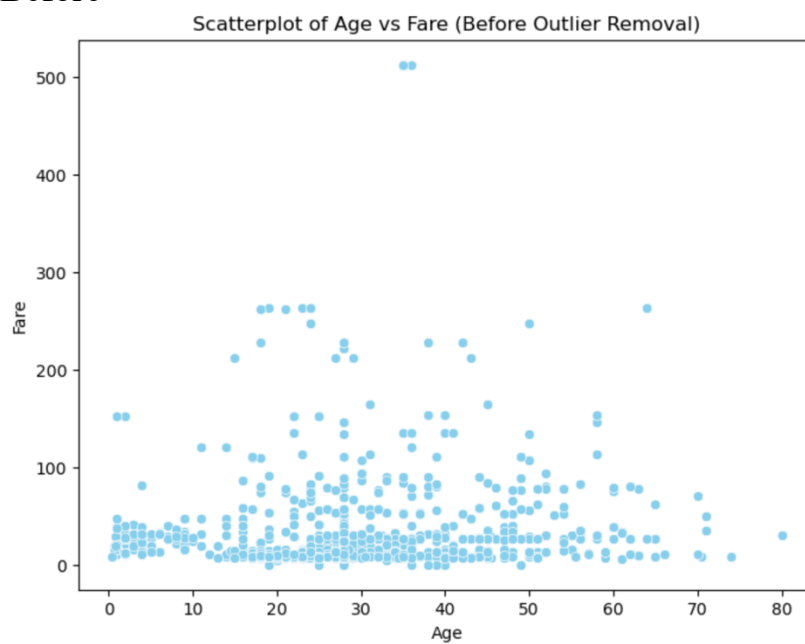


After Removal

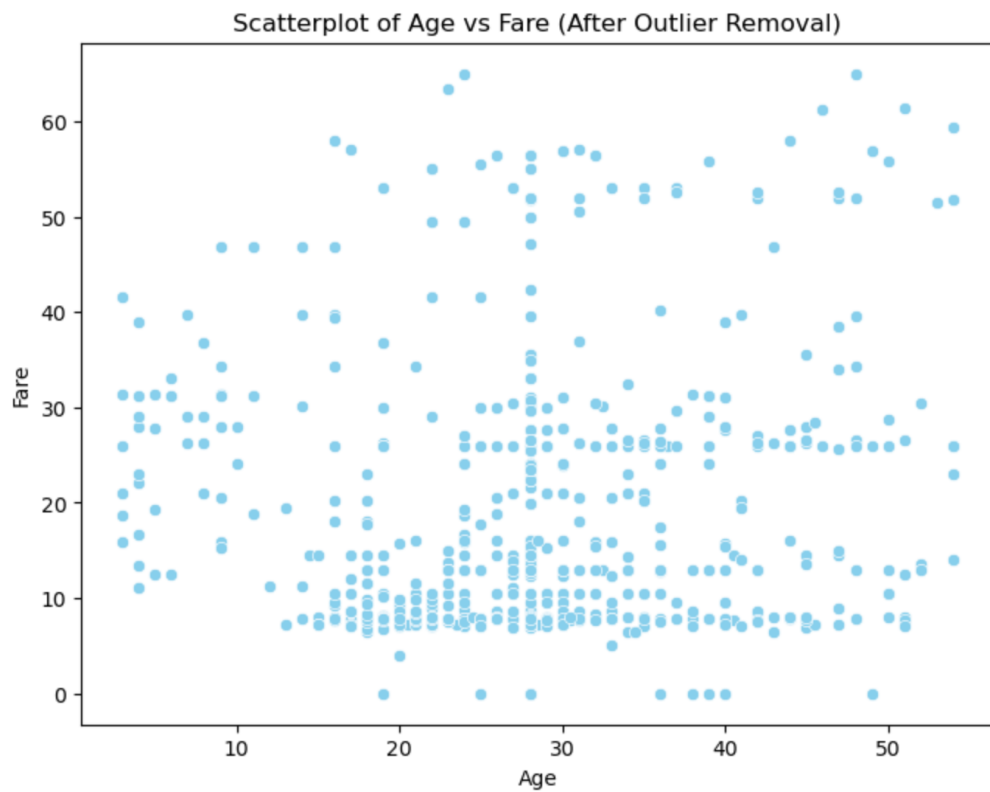


Step 5: Scatterplot (Age vs. Fare)

- Majority cluster between ages 20–40 and fares <100.
- Few high-paying passengers were older and likely from 1st class.
- Before

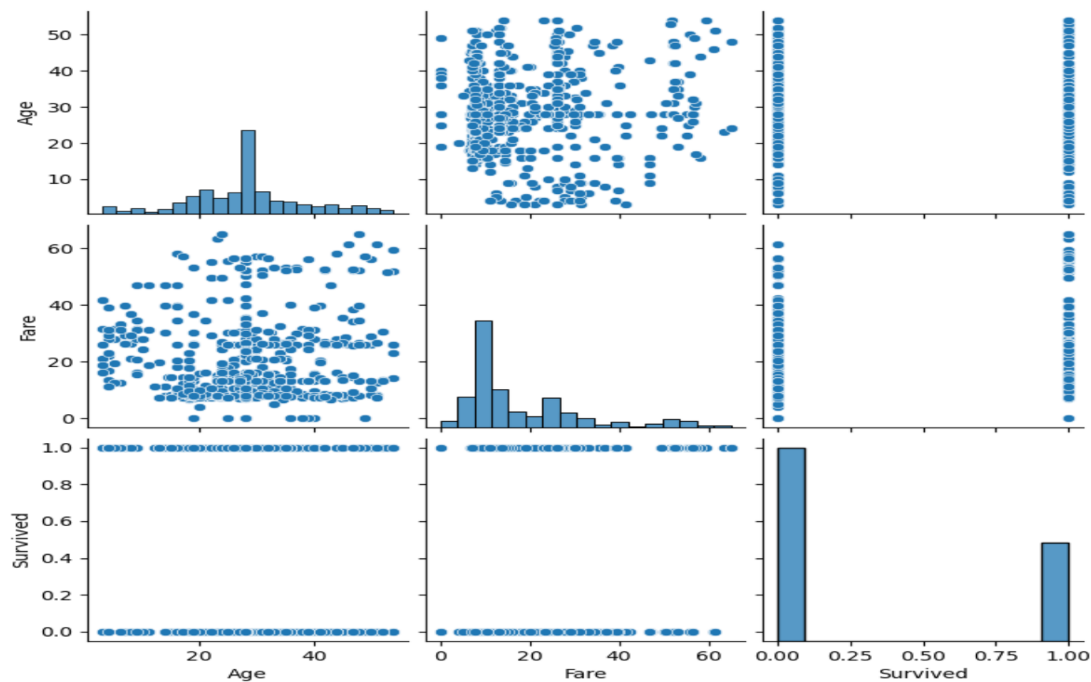


- After



Step 6: Pairplot

- Clear clustering of survived vs not-survived by Age, Pclass, and Fare.
- Survived passengers mostly paid higher fares and were younger.
- Indicates **class and fare had strong influence on survival**.



Step 7: Heatmap (Correlation Matrix)

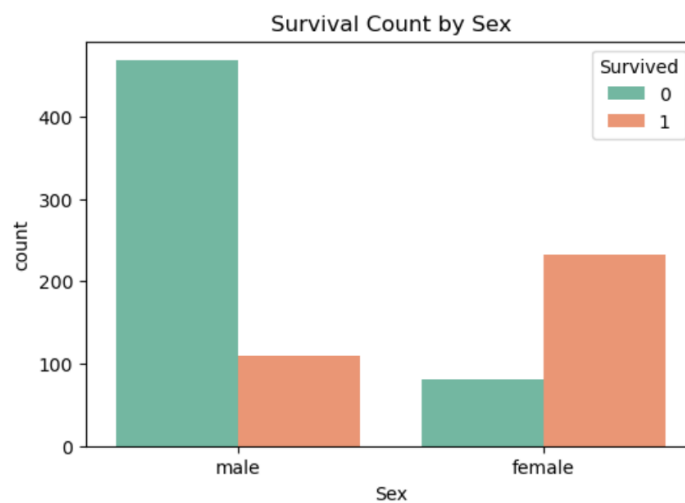
- Positive correlation between **Fare** and **Survived**.
- Negative correlation between **Pclass** and **Survived** (lower class, lower survival).
- **SibSp** and **Parch** had weak positive correlation with survival



Step 8: Countplots (Categorical Analysis)

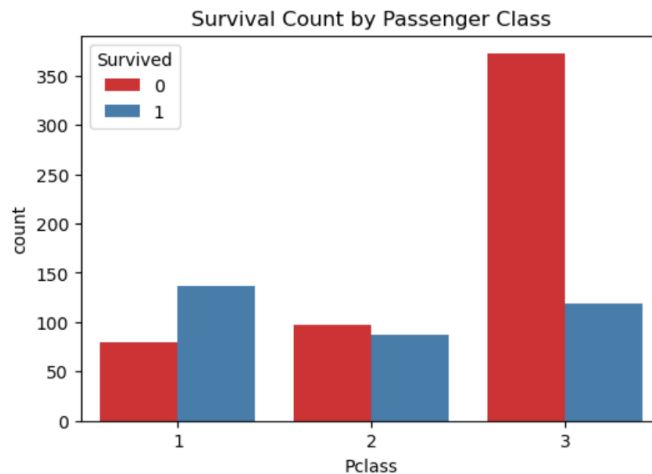
📌 Survived vs Sex

- Females had much higher survival rate than males.



Survived vs Pclass

- **1st class passengers** had the highest survival.
- **3rd class passengers** had the lowest.



Summary of Findings

1. **Females** had a much higher survival rate than males.
2. **Higher-class (1st class) passengers** survived more than 3rd class passengers.
3. Passengers who paid **higher fares** were more likely to survive.
4. **Younger passengers (especially children)** had slightly better chances of survival.
5. Embarkation point also had some influence—**Cherbourg** had the most survivors.
6. **Age and Fare** had outliers, which were handled before visualization.
7. Most people boarded from **Southampton** and were in **3rd class**.